

Coursework Part B:

Locations of Bases for the Welsh Air Ambulance Service

During the second five weeks of the Lent term (Weeks 16 - 20), your task will be to work in small groups in order to investigate the following issue:

A project examining locations of bases for air ambulances, and shift patterns, has recommended reducing the number of bases from four to three and changing the shift patterns, to improve the response to demand. A campaigning group ("Save our Base Welsh Air Ambulance") has questioned these recommendations and asked if there are better ways of improving the service, without reducing the number of bases. Your task is to advise the campaigning group on the following points: i) what are the shortcomings of the modelling work undertaken?, ii) through your own modelling, can you identify any potentially better solutions?

In answering this question, you are expected to use the materials and data provided and apply the analytics skills that you have developed during your MSc so far. As with the previous assignment, the question above is not very precise, and it is up to you to decide on key performance metrics, objectives, decision options, modelling assumptions, etc. to use in your analyses. You will be able to receive support during the timetabled MSCI580 workshop sessions, and will also be encouraged to seek help/advice in other ways, e.g. by posting questions using the MSCI580 Discussion Forum.

The following potentially relevant materials/data are provided on Moodle. They are all included in one PDF document, which is accessible to the public:

- Service Change Pro Forma
- Equality Impact Assessment Tool
- Presentation by the Welsh Air Ambulance Charity. This includes an evaluation of utilisation of current resource, unmet needs, and potential improvement in met demand
- Presentation Overview. This includes more information on the activity of Air Ambulances and Rapid Response Vehicles, and regional demand (and unmet needs) by region and time of day.
- Optima Data Analysis. Optima has conducted the modelling work for the Welsh Air Ambulance Service. This gives more detail on flying hours, and assessment of nine scenarios.
- Comments from Save our Base Welsh Air Ambulance

Your group will be required to submit a report explaining the work that you undertook, by 27 March at 10:00 am, including methods of analyses, main findings and recommendations. The guidance on word length is 3000 to 4000 words. The report will be worth **50%** of the credit for this module.

Important Note: In order to achieve a high mark for this task, you should not only suggest ways of improving the efficiency of the ambulance service, but also show **evidence** (e.g. using relevant statistical analyses or simulation output) that your suggested action(s) will lead to an improved service.